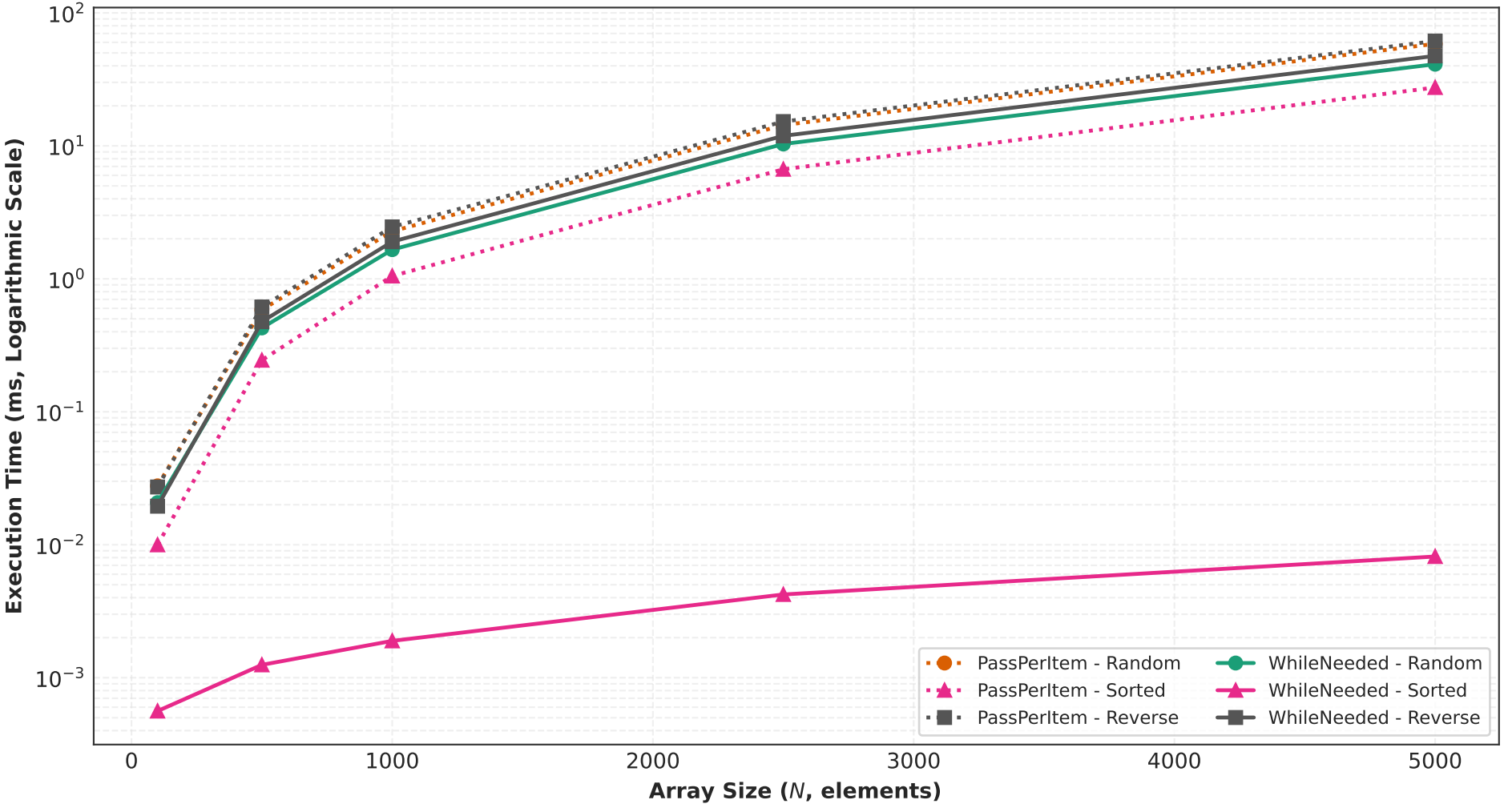
Logarithmic Scaling Landscape: PassPerItem vs. WhileNeeded Across Workloads
Highlighting the Vast Asymptotic Gap ($O(N)$ vs $O(N^2)$) on Pre-Sorted Inputs
Execution Time (ms, Logarithmic Scale)
10^2
10^1
10^0
10^{-1}
10^{-2}
10^{-3}
0
1000
2000
3000
4000
5000
Array Size (N, elements)
PassPerItem - Random
PassPerItem - Sorted
PassPerItem - Reverse
WhileNeeded - Random
WhileNeeded - Sorted
WhileNeeded - Reverse